

Nanosized Cu-MOFs induced by graphene oxide and enhanced gas storage capacity†

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Various MOFs with tailored nanoporosities have recently been developed as potential storage media for CO₂ and H₂. The composites based on Cu-BTC and graphene layers were prepared with different percentages of graphene oxide (GO). Through the characterization analyses and gas adsorption experiments, we found that the nano-sized and well-dispersed Cu-BTC induced by the incorporation of GO greatly improved the carbon dioxide capture and hydrogen storage performance of the composites. The materials obtained exhibited about a 30% increase in CO₂ and H₂ storage capacity (from 6.39 mmol g⁻¹ of Cu-BTC to 8.26 mmol g⁻¹ of CG-9 at 273 K and 1 atm for CO₂; from 2.81 wt% of Cu-BTC to 3.58 wt% of CG-9 at 77 K and 42 atm for H₂). Finally, the CO₂/CH₄ and CO₂/N₂ selectivities were calculated according to single-component gas sorption experiment data.

Introduction

Nowadays, energetic needs are mainly met by fossil fuels with concomitant pollutant emissions responsible for global warming. CO₂ is the main greenhouse gas, so adsorption and separation of CO₂ are of utmost importance for energy development.¹ Simultaneously, hydrogen has been proposed as a clean and efficient energy carrier. However, besides the difficulties related to hydrogen production, high-capacity storage is still to be achieved.² Porous materials based on adsorption processes can be used for carbon dioxide capture and hydrogen storage.^{3–5}

Broader context

Metal–organic frameworks (MOFs) have emerged as a new class of hybrid nanoporous materials. At the same time, graphene and graphene-based materials have recently received attention owing to the very unique structural properties of these one-atom-thick sheets. In the present article, the graphene oxide@Cu-based MOF porous composites contain nano-sized and well-dispersed Cu-BTC crystallites on the layers of GO, and exhibited about a 30% increase in CO₂ and H₂ storage capacity. The enhanced gas uptake capacity of composites has potential applications in the field of energy and environment.

Recently, MOFs have been evaluated as promising CO₂ and H₂ storage media as MOFs exhibit exceptionally high surface areas and tunable pore size. Among the reported MOFs, Cu-BTC [Cu₃(BTC)₂(H₂O)₃; BTC = benzene-1,3,5-tricarboxylate] is thermostable and exhibits high porosity, suggesting that Cu-BTC is a promising gas storage medium. The polymer framework of Cu-BTC is composed of dimeric cupric tetra-carboxylate units (Fig. S1†). Each metal completes its pseudooctahedral coordination sphere with an axial aqua ligand opposite to the Cu–Cu vector.⁶

Graphene and graphene-based materials have recently received attention owing to the specific and very unique structural, mechanical, optical and electronic properties of these one-atom-thick sheets.^{7,8} GO has a layered structure and a non-stoichiometric chemical composition. According to the Lerf–Klinowski model of GO,⁹ the basal plane of GO is decorated with functional groups such as OH (hydroxyl group) and C–O–C (epoxy group), whereas carboxylic groups are mainly found at the edges. Recently, GO has been used to prepare various nanocomposites with enhanced properties.^{10–12} Recently, several groups have reported a few studies on the synthesis and applications of graphene-based MOFs, due to their high porosity, adsorption capacity for specific gases and electric properties.^{13–18}

In the work, we prepared the hybrid composites based on Cu-BTC and graphene oxide layers (denoted as CG). We discussed

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the influence on the Cu-BTC crystal size induced by the incorporation of GO into the composites, and discovered that nanosized Cu-BTC crystallites are well dispersed on the graphene oxide layers.

Nanocrystallized and well-dispersed Cu-BTC has greatly improved the CO₂ capture and H₂ storage performance of the composites. The CO₂/CH₄ and CO₂/N₂ selectivities were calculated according to single-component gas sorption experiment data. Therefore, the GO-MOF composites have great potential applications in the field of energy.

Experimental section

Materials

Graphite oxide was synthesized by oxidation of graphite using an improved Hummers method.¹⁹ Cu-BTC (HKUST-1) was prepared by mixing copper nitrate trihydrate (1 mmol) and 1,3,5 benzenetricarboxylic acid (0.67 mmol) in a mixture of 12 ml DMF, 12 ml ethanol and 8 ml deionized water followed by stirring for 5 min. The reaction was heated to 80 °C for 24 h in a Teflon-lined 40 ml Parr pressure vessel. The resulting product was kept in a desiccator and is referred to as Cu-BTC.

The preparation of composites was done by adding MOF precursors to dispersions of graphene oxide which was then exfoliated by sonicating graphene oxide in deionized water sufficiently and performing the same procedure as in the preparation of Cu-BTC. The added GO consisted of 3, 9 and 15 wt% of the material weight, we assume that Cu-BTC was formed in 100% yield. The composites are referred to as CG-3, CG-9 and CG-15 respectively.

Characterization

Powder XRD experiments were conducted on a PANalytical X-ray diffractometer at 40 kV, 40 mA, with a scan speed of 5° min⁻¹ and a step size of 0.02° in 2 θ , using Cu K α radiation. High resolution transmission electron microscopy (HRTEM) was performed on a Tecnai G2 F30 instrument. Scanning electron microscopy (SEM) images of the morphology of Cu-BTC crystals were taken on a JSM-6360LV SEM operating at an accelerating voltage of 20 kV. Thermogravimetric analysis (TGA) was carried out on a Cahn Thermax 500 instrument. Fourier transform infrared (FT-IR) spectra were recorded on a Nicolet 380 spectrometer, and the samples for FT-IR were prepared by mixing dried materials with KBr and then pressing into wafers.

Gas adsorption measurement

A Low-pressure gas adsorption measurements. Adsorption isotherm studies for materials were performed volumetrically as a function of the gas relative pressure by using an Autosorb 1 from Quantachrome. Samples were weighed precisely after outgassing at 150 °C for 6 hours. The low temperature of the sample for N₂ adsorption experiments was stabilized and controlled with the help of a refrigerated bath of liquid nitrogen (77 K). High temperature isotherms of the sample for CO₂, CH₄ and N₂ adsorption (273 K) were measured in an ice-water bath.

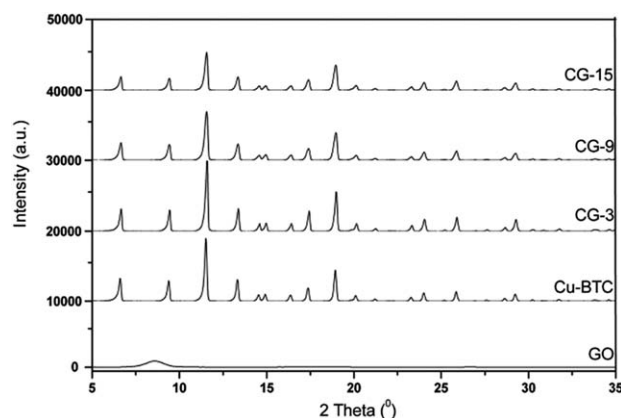


Fig. 1 X-ray diffraction patterns of the parent materials and the composites.

B High-pressure hydrogen adsorption measurements.

High-pressure hydrogen uptake was measured with a commercial pressure-composition isotherm (PCI) unit provided by Advanced Materials Co. (USA). Highly purified hydrogen (99.9995% purity) was used as the adsorbent. Prior to sorption measurements, materials were activated under vacuum at 373 K to remove any physisorbed water or volatile impurities. Leak testing was also performed during each measurement. The sample was run at 77 K and 42 atm.

Results and discussion

XRD patterns

Fig. 1 provides the X-ray diffraction patterns of the parent materials as well as the composites, CG-3, CG-9 and CG-15. The good agreement between the pattern of Cu-BTC and the one reported in the literature demonstrated a successful synthesis procedure.²⁰ The similar diffraction patterns of the composites to Cu-BTC indicate the existence of the well-defined MOF units and thus prove that GO did not prevent the formation of linkages between the copper dimers and the organic bridges, but the intensity of the peaks is decreased with increasing GO content. On the other hand, a single peak around 2 θ of 8.9° observed for GO reveals a distance between the carbon layers of 9.9 Å as determined by Bragg's law. It is interesting to notice that the peak of GO is not found for the composites. One reason should be that the XRD patterns of GO show a low intensity broad peak compared to the high intensity sharp peaks of Cu-BTC, and the content of GO in the composites is relatively low (3 wt%, 9 wt% and 15 wt%). Another explanation is likely due to ascribed exfoliation/high dispersion of GO because of the presence of polar solvents (in particular DMF) which are known to disperse GO very well.²¹ Cu-BTC crystals and exfoliated graphene oxide layers can be observed in the HRTEM images.

HRTEM image analysis

The wrinkled graphene oxide layers are shown in Fig. 2a. The size of Cu-BTC obtained as cube-shaped crystals was around 10 μ m (Fig. S2†), which is consistent with the previous report.²² Well-defined graphene-based layers with Cu-BTC crystallites are

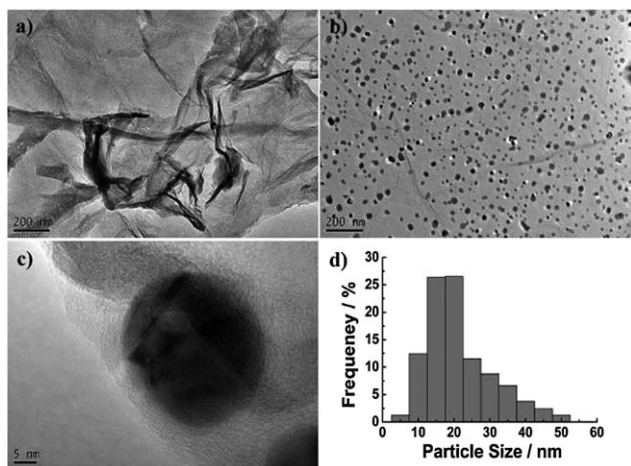


Fig. 2 HRTEM images for the wrinkled graphene oxide layers (a), CG-9 (b) and a single Cu-BTC particle on the GO layer of CG-9 (c) and (d) the histogram of the Cu-BTC particles size distribution.

observed in the HRTEM image of composite CG-9 (Fig. 2b). It is interesting that the shape of many Cu-BTC crystals is hexagonal, as shown in Fig. 3. And it is obvious that Cu-BTC shows a hexagonal-shaped unit cell viewed down the [111] direction (Fig. S3†), which belongs to cubic crystal system.⁶ Camille Petit *et al.* reported that MOF “blocks” attach to a graphene layer by the interaction between metal sites of MOFs and epoxy groups on GO.^{13,14,17} Based on the above mentioned facts, it can be concluded that epoxy groups on the GO layers act as seed sites for Cu-BTC crystallization, and it is plausible to assume that the [111] direction is the preferential growth orientation of Cu-BTC crystals. As shown in the size distribution histogram (Fig. 2d), most of the Cu-BTC crystallites are in the range of 10–40 nm. Among them, 75% of Cu-BTC crystallites are in the range of 10–25 nm. Therefore, epoxy groups on the GO layers prevent aggregation of crystallites and increase dispersion, which led to the formation of nanosized and well-dispersed Cu-BTC crystallites. In fact, the surface of GO is not smooth. Cristina Gomez-Navarro *et al.* reported that the layers of RGO (RGO means Reduced Graphene Oxide and GO is prepared according

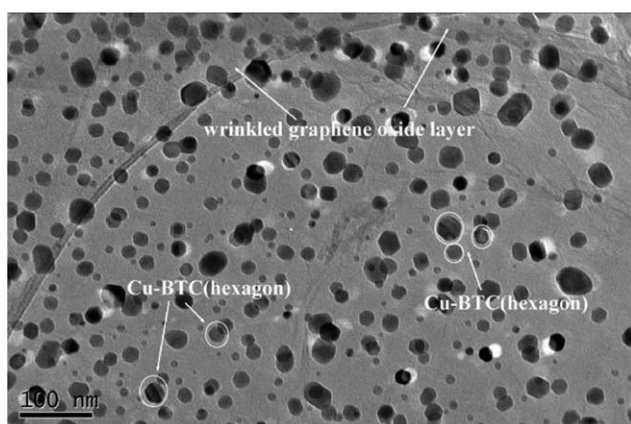


Fig. 3 HRTEM images of the composite CG-9.

to a chemical oxidation method) were found to comprise defect-free graphene areas with sizes of a few nanometers interspersed with defect areas dominated by clustered pentagons and heptagons.²³ In order to investigate the interaction between the GO and Cu-BTC, the adsorption energy of Cu atoms on the GO surface is calculated using density functional theory (DFT) implemented in the Vienna *ab initio* simulation package (VASP). These calculation results show that the adsorption of Cu atoms on the GO defect with a binding energy of 15.12 eV is more favorable than those on the –OH (3.92 eV) and –O– groups (1.60 eV) of the GO surface (see Fig. S4†). Therefore, Cu atoms bonded with C atoms on the defects of GO are almost coordinatively saturated, and then Cu-BTC nanocrystals will stop growing when it reached the defects of the GO. Furthermore, there are two reasons for the formation of nanosized and well-dispersed Cu-BTC on the layers of GO. First, epoxy groups on the GO layers act as seed sites for well-dispersed Cu-BTC crystallization. Second, the defects of GO prevent the growth of Cu-BTC nanocrystals. And then, nanosized Cu-BTC can greatly favor to take full advantage of the porosity, which was confirmed by N₂ adsorption experiments. Moreover, it is known that electron beam irradiation can cause the collapse of Cu-BTC and can thus prevent any visualization of its lattice structure.²⁴ However, the pattern of Cu-BTC can be observed in the HRTEM image of CG-9 (Fig. 3). It is possible that GO distorted graphene-based layers present in the composites helped the MOF to retain its crystalline structure by dissipating the electrostatic charges. The phenomenon was also observed by Yang *et al.* with their MOF-5/CNTs composites.²⁵

Gas adsorption analysis

To evaluate the porosity of the materials, nitrogen adsorption measurements were carried out at 77 K under 1 atm, and the isotherms are plotted in Fig. 4. Table 1 lists the BET specific surface areas of Cu-BTC, CG-3, CG-9 and CG-15, as determined from N₂ adsorption isotherms. The porosity of GO can be negligible. Cu-BTC has a BET-SSA of 1305 m² g^{−1} which is in the range of the values reported in references under hydrothermal

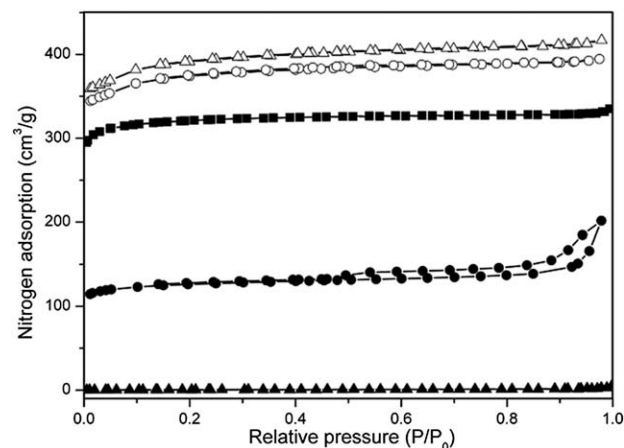


Fig. 4 Nitrogen adsorption isotherms of GO (▲), Cu-BTC (■), CG-3 (○), CG-9 (△) and CG-15 (●).

Table 1 Summary of adsorption properties of Cu-BTC, CG-3, CG-9 and CG-15^a

Sample	S_{BET} ($\text{m}^2 \text{g}^{-1}$)	V_t ($\text{cm}^3 \text{g}^{-1}$)	H_2 uptake (wt%)	CO_2 uptake (mmol g^{-1})	Average particle size (nm)
Cu-BTC	1305	0.5182	2.81	6.39	—
CG-3	1470	0.6103	3.05	7.94	25.6
CG-9	1532	0.6448	3.58	8.26	21.6
CG-15	500	0.3119	1.42	2.97	64.5

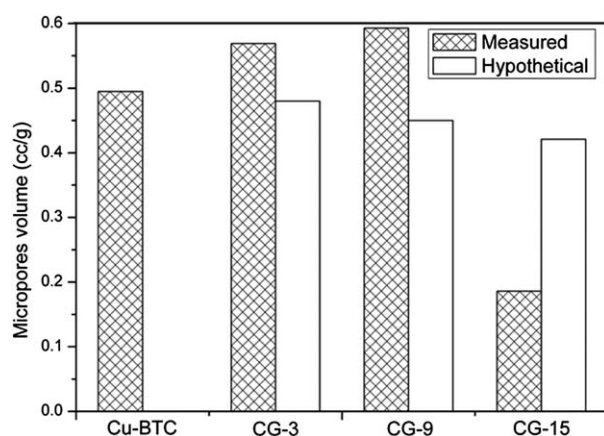
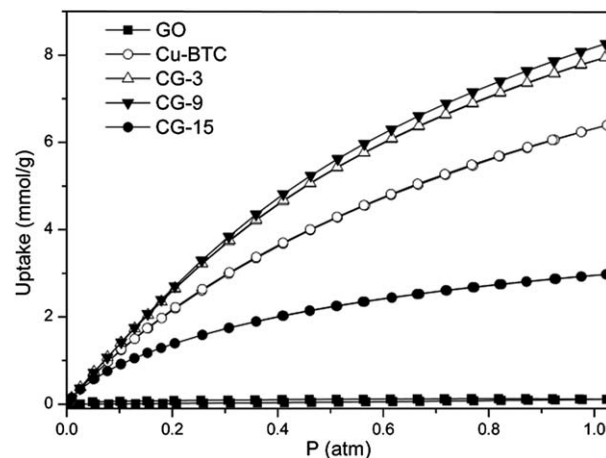
^a H_2 uptake at 77 K and 42 atm; CO_2 uptake at 273 K and 1 atm; average particle sizes of Cu-BTC nanocrystals are measured according to the TEM images of composites, as shown in Fig. S5.

conditions.^{6,26} Cu-BTC and the composites exhibit a type I isotherm. This indicates the predominant microporous character of these compounds. This is expected since Cu-BTC is characterized by square channels of 9 Å by 9 Å.⁶ Therefore, it is significative to compare the measured values of the composites' micropore volumes with the hypothetical ones, which represent the parameters of the porous structure of the physical mixture of GO and Cu-BTC (Fig. 5). These hypothetical micropore volumes were calculated by taking into account the percentage of each component in the composites and the micropore volumes of GO and Cu-BTC alone. Eqn (1) provides the details of the calculation:

$$X_n = X_{\text{GO}} \times \text{wt}\%_{\text{GO}} + X_{\text{Cu-BTC}} \times \text{wt}\%_{\text{Cu-BTC}} \quad (1)$$

where " X_n " is the micropore volume of composite CG- n to be determined, " X_{GO} " and " $X_{\text{Cu-BTC}}$ " are the micropore volumes of GO and Cu-BTC respectively, and " $\text{wt}\%_{\text{GO}}$ " and " $\text{wt}\%_{\text{Cu-BTC}}$ " are the weight percentages of GO and Cu-BTC in CG- n . It is noteworthy that the measured values of the composites with the lower GO contents (CG-3 and CG-9) are higher than those calculated for the physical mixture.

The porosity of MOFs, which makes them particularly attractive, is often not fully utilized. The "incorporation" of GO in the composites was expected to prevent aggregation and favor the dispersion of nanosized MOF crystallites, confirmed

**Fig. 5** Measured and "hypothetical" micropore volumes of the porous structure for Cu-BTC and the composites.**Fig. 6** CO_2 adsorption isotherms of Cu-BTC, CG-3, CG-9 and CG-15 at 273 K.

by the HRTEM images. The increased micropore volumes and BET-SSAs of CG-3 and CG-9 can be attributed to the well-dispersed MOF crystallites which improve the porosity and accessibility of the MOF network for the adsorbent. Furthermore, Hailing Guo and Hongjie Zhang reported that the decrease in crystal size from bulk to nanoscale for MOFs has a significant effect on the surface area.²⁷ As shown in Table 1, it is obvious that CG-9 has the least size and the highest surface area. However, the opposite trend is observed for composite CG-15. First, the micropore volume of CG-15 is reduced by the more neglectable pore volume of GO. Second, excessive GO may not be completely dispersed, which will cause too much distortion in the structure of the composites, confirmed by the decreased intensities of the XRD peaks with increasing GO content for the composites. Third, too much graphene oxide will baffle N_2 to reach MOF crystals, confirmed by the significant reduction of the total pore volume for CG-15.

The isotherms for CO_2 on GO, Cu-BTC, CG-3, CG-9 and CG-15 are given in Fig. 6. The CO_2 storage capacities of Cu-BTC exhibit a steep rise at low pressure, and reach a maximum of 6.39 mmol g^{-1} at 273 K and 1 atm, higher than most of MOF materials at the same pressure.²⁸ When GO was incorporated into the Cu-BTC crystals, the CO_2 storage capacities of CG-3 (7.94 mmol g^{-1}) and CG-9 (8.26 mmol g^{-1}) were enhanced by 24.3% and 29.3% respectively, higher than that of Cu-BTC. The 8.26 mmol g^{-1} of CG-9 is among the highest uptake capacity ever reported for porous MOF-based materials under the same conditions. A plot of molar uptake as a function of pore volume reveals a nearly linear correspondence (Fig. S6†). This illustrates that the CO_2 adsorption properties are similar to those of N_2 for CG-3 and CG-9, caused by improved porosity and accessibility of the well-dispersed nanosized MOF crystallites for CO_2 . On the other hand, too much GO in CG-15 resulted in a lower CO_2 capacity than that of Cu-BTC.

The hydrogen storage properties of the Cu-BTC and composites were measured. The H_2 adsorption behaviors of samples at 77 K and 42 atm are shown in Fig. 7. Cu-BTC exhibited a H_2 uptake of 2.81 wt% at 42 atm, consistent with the range of uptakes reported in the literature.^{29,30} The H_2 storage

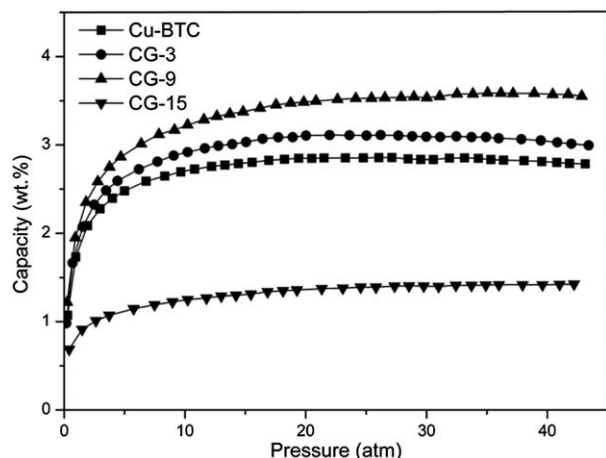


Fig. 7 H_2 adsorption isotherms of Cu-BTC, CG-3, CG-9 and CG-15 at 77 K and 42 atm.

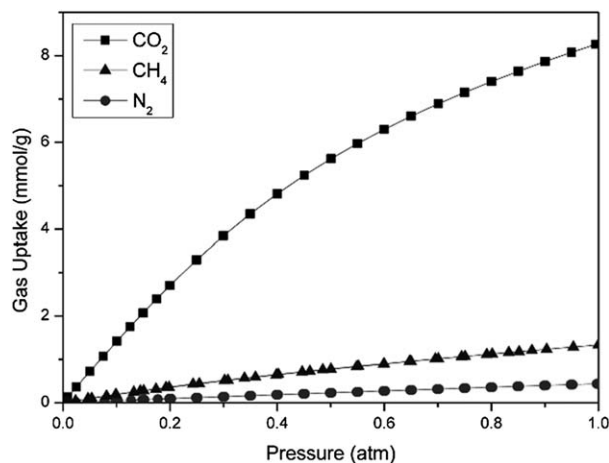


Fig. 8 Adsorption isotherms of CG-9 for CO_2 , CH_4 and N_2 at 273 K.

capacities of CG-3 (3.05 wt%) and CG-9 (3.58 wt%) were improved by 8.5% and 27.4% respectively, higher than that of Cu-BTC.

To examine the separation capability of Cu-BTC and the composites for various gases in an admixture, single-component gas sorption experiments were carried out on CO_2 , CH_4 and N_2 at 273 K and up to 1 atm (adsorption isotherms of CG-9 are shown in Fig. 8 and the others are shown in Fig. S9†).³¹ The initial slopes of CO_2 , CH_4 and N_2 adsorption isotherms were calculated, and the ratios of these slopes were used to estimate the adsorption selectivity for CO_2 over CH_4 or N_2 (Fig. S10†). The

Table 2 CO_2/CH_4 and CO_2/N_2 selectivities of Cu-BTC, CG-3, CG-9 and CG-15^a

Selectivity	Cu-BTC	CG-3	CG-9	CG-15
CO_2/CH_4	6.6 : 1	6.5 : 1	7.3 : 1	4.6 : 1
CO_2/N_2	41 : 1	36 : 1	30 : 1	27 : 1

^a CO_2 , CH_4 and N_2 uptakes at 273 K and 1 atm; the initial slope calculation of Cu-BTC, CG-3, CG-9 and CG-15 for CO_2 , CH_4 and N_2 is shown in Fig. S10.

CO_2/CH_4 and CO_2/N_2 selectivities of Cu-BTC, CG-3, CG-9 and CG-15 are shown in Table 2.

Conclusions

In conclusion, nanosized and well-dispersed Cu-BTC crystallites are obtained in the composites, due to crystal nucleation anchored by epoxy groups on the “incorporation” of GO layers. Nanosized Cu-BTC enhanced the gas adsorption properties of the composites, especially CG-9. Composite CG-9 can adsorb $8.26 \text{ mmol g}^{-1} CO_2$ at 273 K under 1 atm and 3.58 wt% H_2 at 77 K under 42 atm. The CO_2/CH_4 and CO_2/N_2 selectivities were calculated according to single-component gas sorption experiment data. These very useful results suggest a new approach to increase MOF gas storage performances and great potential applications in the field of energy and environment.

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